THERMAL CONTROL SUBSYSTEM

Temperature under control.

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A proposed monitoring and shutdown node for the modular 2 DOF robotic-arm motor controller.

SENSE	CONDITION	DECIDE	PROTECT
3 NTC probes	MCP6002 buffers	PIC18F57Q43	Shared FAULT_N

Monitor the pitch motor driver, roll motor driver and local 5 V regulator. Each probe uses a 10 kOhm pullup to 5 V and an NTC to ground. Temperature rises as the sensed voltage falls.

Candidate hardware

Part / interface	Proposed role
3 x Vishay NTCLE100E3103JB0 [1]	10 kOhm at 25 deg C; B25/85 = 3977 K. R25 +/-5%; beta +/-0.75%.
2 x MCP6002 [2]	Three unity-gain buffer channels; define the unused amplifier. Each ADC input: 1 kOhm series + 100 nF to ground.
PIC18F57Q43 Curiosity Nano [3,4]	12-bit ADC on RA0 / RA1 / RA2; local status, threshold checks and USB serial diagnostics.
L7805ACV + protected 9 V input [7]	Independent 5 V logic rail with fuse, bypass capacitors and power indication.
P2N2222A / optional IRLZ44NPBF [5,6]	Open-collector fault sink / local fan power switch. Fan stays on the fused 9 V branch.

Proposal status. Parts, probe placement, thresholds and wiring remain subject to team review and bench verification. No assembled-board, calibration, firmware-completion or measured-performance claim is made.

CALCULATED EXAMPLES

From resistance to a decision.

Use the ADC reference from the same 5 V rail as the divider to make the ideal measurement ratio ratiometric. These values use a beta-model approximation; calibration and the manufacturer curve are still needed.

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R(T) = 10,000 exp[3977 (1/(T + 273.15) - 1/298.15)]
V_SENSE = VDD x R(T) / (10,000 + R(T))
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T in deg C; R in Ohm. Guard zero/full-scale readings before conversion.

State example	Temperature	Sense at 5 V	12-bit code*
Manual rearm allowed	<=45 deg C	1.5092 V	1236
Warning / optional fan on	>=50 deg C	1.3135 V	1076
Latched shutdown request	>=60 deg C	0.9880 V	809

*Rounded display approximation: code = 4095 x V_SENSE / VDD. At 25 deg C: 2.500 V, code about 2048. Settings above are illustrative; actual motor-driver and regulator limits must set the final thresholds.

A preliminary 5 V power budget

Load / allocation	Current
PIC/Nano operating allowance - unverified	100.00 mA
3 NTC branches + 4 amplifier channels	2.18 mA
3 status LEDs + fault base + standalone pullup	10.00 mA
Regulator minimum load + supporting circuit allowance	15.00 mA
Planned total / with 25% margin	127.18 mA / 158.98 mA

0.160 A
rounded logic-rail allocation

0.694 W
nominal regulator screening loss

Loss estimate: (9 - 5) x 0.160 + 9 x 0.006. Include actual adapter tolerance, regulator ground current, ambient, heatsinking and fan startup before sizing the source. The optional fan adds a provisional 0.20 A at 9 V; this is not a verified part maximum. [7]

INTEGRATE. THEN VERIFY.

Make the fault path explicit.

Bus pin 5: FAULT_N, active low. P2N2222A collector connects through an isolation jumper to pin 5; emitter goes to common ground on pin 8. A GPIO drives its base through 10 kOhm; add 100 kOhm base-to-emitter. Install only one 10 kOhm pullup across the complete bus. Never drive this shared line high. [5]

Startup limitation. At MCU reset the base pulldown turns the fault transistor off. This circuit alone does not guarantee shutdown on MCU failure, reboot or complete local power loss. Drive nodes must begin inhibited and require valid-node checks plus deliberate arming. A serious thermal or sensor fault remains latched until cooling and manual rearm.

Keep the ribbon allocation intact: pin 6 is roll-position feedback, pin 7 is joint-load feedback and pin 8 is ground. No supply power is shared. Temperature telemetry can stay local over USB; another inter-node method needs agreement. The optional 9 V fan does not load the 5 V regulator.

Bench verification plan

01 Power & startup
Verify the protected input, 5 V rail, Nano VTG/VOFF wiring and disabled startup. Check regulator heating at the planned load. [4,7]

02 Sensor chain
Substitute known resistors; check all ADC channels, acquisition timing, open/short detection, noise and final thermal attachment. [1-3]

03 Thresholds & cooling
Use a reference thermometer and controlled heating. Calibrate, check warning/trip/rearm behavior, and test the selected fan startup current.

04 Shared shutdown
Assert a fault, scope pin 5 to motor cutoff, check latch behavior and verify deliberate rearming. The team <20 ms target remains unmeasured.

Primary references - clickable links

[1] Vishay NTCLE100E3
[2] Microchip MCP6002
[3] Microchip PIC ADC / acquisition
[4] Curiosity Nano hardware guide
[5] onsemi P2N2222A
[6] Infineon IRLZ44N
[7] ST L78 regulator family
[8] TI LM393 comparator options
[9] Analog Devices TMP36 alternative

References 8-9 are comparison options, not installed parts. Thermal accuracy, final fuse value, probe insulation, fan compatibility and independent hardware protection remain open design items.